Graph Theroy

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Chapter 1

Introduction to Graph Theory

1.1 What is Graph Theory?

At its core, **graph theory** is the study of relationships. Whenever we have a collection of objects and connections between them, we can use a graph to model, analyze, and solve problems involving that structure.

Unlike the graphs you might recall from introductory algebra (like $y = x^2$ or sine waves), a graph in discrete mathematics is a structural map. It consists of individual points and the lines that connect them.

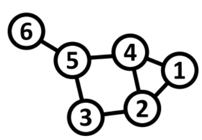
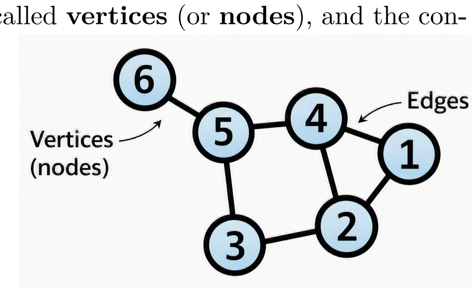


Figure 1.1: A graph with 6 vertices and edges *(Image Credit: Wikipedia)*

The objects are represented as points called **vertices** (or **nodes**), and the connections between them are called **edges**.



Graphs give us a unified mathematical language to represent networks. Whether we are analyzing social networks, computer communication grids, or transit maps, graph theory provides the core framework to understand how systems are connected.

1.2 Historical Foundations

To appreciate graph theory, it helps to understand the historical puzzles that sparked its development.

1.2.1 The Seven Bridges of Königsberg (1736)

Graph theory was born in the Prussian city of Königsberg (now Kaliningrad, Russia). The city was divided by the Pregel River, creating two large islands connected to each other and to the mainland riverbanks by seven bridges.

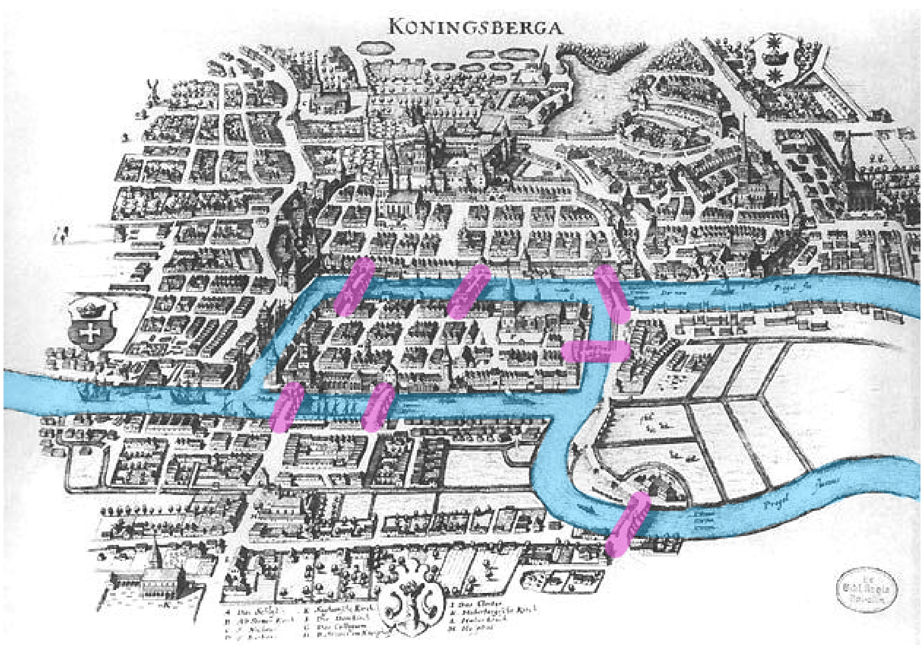


Figure 1.2: Image Source: Wikipedia

The citizens posed a classic recreational puzzle: *Is it possible to take a walk through the city in such a way that you cross every single one* *of the seven bridges exactly once, and end up back where you started?*

In 1736, the Swiss mathematician **Leonhard Euler** solved this problem. He realized that the specific distances, land shapes, and physical geography did not matter; only the **connections** mattered. Euler simplified the map by replacing each landmass with a point (vertex) and each bridge with a line (edge).

By abstracting the geography into a mathematical structure, Euler proved that such a walk was impossible. In doing so, he laid the foundation for topology

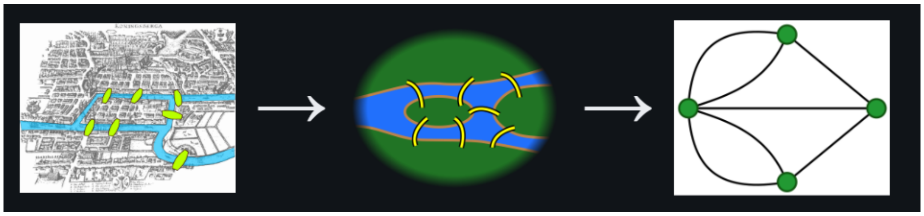


Figure 1.3: Image Credit: Wekipdea

and graph theory.

1.2.2 Hamilton's Icosian Game (1859)

Over a century later, in 1859, Irish mathematician **Sir William Rowan Hamilton** created a commercial puzzle called the *Icosian Game*.

The game used a solid wooden regular dodecahedron—a 12-faced polyhedron with 20 corners. Each of the 20 corners was labeled with the name of a famous world city. The objective was to find a route along the edges of the dodecahedron that visited **every city exactly once** and returned to the starting point.

While Euler's bridge problem focused on visiting every *edge* once, Hamilton's game focused on visiting every *vertex* once. These two historical problems led directly to two major areas of graph theory: **Eulerian paths** and **Hamiltonian cycles**.

1.3 Modern Applications

Today, graph theory is an essential foundation for computer science, data analytics, and operational engineering.

- **Social Networks**: Platforms model users as vertices and friendships or followers as edges. Graph theory algorithms help identify tight-knit communities, influential users, and recommendation loops.
- **Transportation Networks**: Airline routes, train networks, and road grids are modeled as graphs. GPS platforms rely on these models to compute optimal routes.
- **The Web & PageRank**: The World Wide Web is a massive graph where web pages are vertices and hyperlinks are directed edges. Early

Figure 1.4: Porteat of Sir William Rowan Hamilton (Image Credit: Wikipedia)

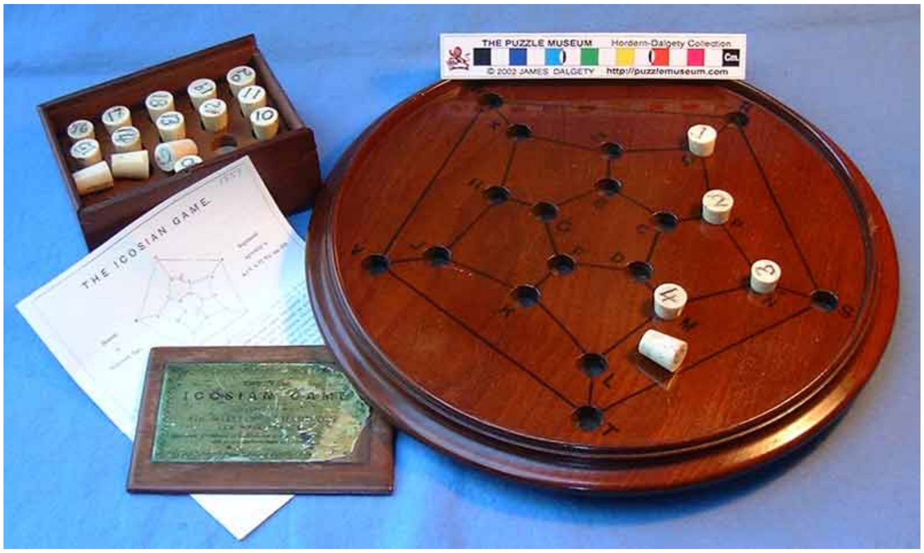


Figure 1.5: The Icosian Game(Image Credit: puzzlemuseum)

search engines transformed web searches by using graph structure to rank relevance based on link density.

1.4 Formal Definitions and Basic Structures

Let us establish the formal mathematical definitions used throughout graph theory.

1.4.1 Mathematical Definition of a Graph

A **graph** G is a pair $G = (V, E)$, where:

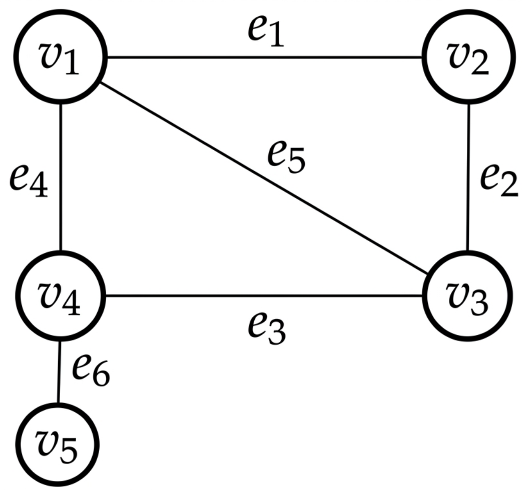
- V is a non-empty set of **vertices** (or nodes).
- E is a set of **edges**, where each edge $e \in E$ is associated with an unordered pair of vertices $\{u, v\} \subseteq V$.

If an edge e connects vertices u and v, we write $e = (u, v)$ or $e = uv$. We say that e is **incident** with u and v, and that u and v are **adjacent** (or neighbors).

1.4.1.1 Example

Example 1.1. Consider a graph $G = (V, E)$ defined by:

- $V = \{v_1, v_2, v_3, v_4, v_5\}$
- $E = \{e_1, e_2, e_3, e_4, e_5, e_6\}$ where:
- $e_1 = (v_1, v_2)$
- $e_2 = (v_2, v_3)$
- $e_3 = (v_3, v_4)$
- $e_4 = (v_4, v_1)$
- $e_5 = (v_1, v_3)$
- $e_6 = (v_4, v_5)$



In this graph:

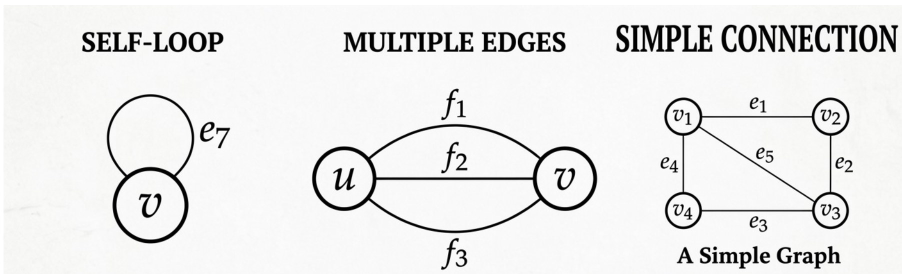
- v_1 and v_3 are adjacent because $e_5 = (v_1, v_3)$ connects them.
- v_2 and v_5 are **not** adjacent because there is no edge directly connecting them.

1.4.2 Loops, Multiple Edges, and Simple Graphs

Not all graphs are strictly constructed with single lines between distinct points.

- **Self-loop**: An edge that joins a vertex to itself (e.g., $e = (v_1, v_1)$).
- **Multiple Edges (Parallel Edges)**: A collection of two or more edges that connect the exact same pair of distinct end vertices.

- **Simple Graph**: A graph that contains **no self-loops and no multiple edges**.



Unless specified otherwise, standard graph theory problems assume a graph is **simple**.

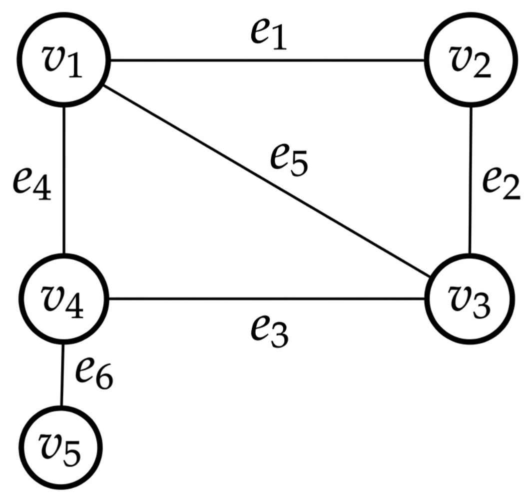
1.5 Vertex Degrees and Handshaking

1.5.1 Vertex Degree

The **degree** of a vertex v, denoted by $d(v)$, is the number of edges incident to v. For simple graphs, it is equal to the number of neighbors connected to v.

(Note: If a vertex has a self-loop, that loop contributes 2 *to the vertex's degree, as it touches the vertex twice.)*

1.5.1.1 Example



Example 1.2.
Using Example 1.1 above:

- $d(v_1) = 3$ (connected by e_1, e_4, e_5)
- $d(v_2) = 2$ (connected by e_1, e_2)
- $d(v_3) = 3$ (connected by e_2, e_3, e_5)
- $d(v_4) = 3$ (connected by e_3, e_4, e_6)
- $d(v_5) = 1$ (connected by e_6)

Summing all degrees: $3 + 2 + 3 + 3 + 1 = 12$. Notice that the total number of edges $|E|$ is 6. The sum of degrees (12) is exactly twice the number of edges (2×6).

1.5.2 The Handshaking Lemma

This observation holds true for all graphs.

Theorem 1.1 (Handshaking Lemma)**.** *Let $G = (V, E)$ be any graph. Then the sum of the degrees of all vertices is equal to twice the number of edges:*

$$\sum_{v \in V} d(v) = 2|E|$$

Proof. Intuition: